Practical – 8

Practical 8: MySQL Schema Design, ER Model, PDO Connectivity, and Prepared Statements

Problem Definition: Design the database for StudentHub with students, events and registrations. Create PHP database connection using MYSQL statements.

Theory :

1. Is the schema normalized properly?

Yes. The database is divided into separate **Students, Events, and Registrations** tables, which reduces duplicate data and keeps the data organized.

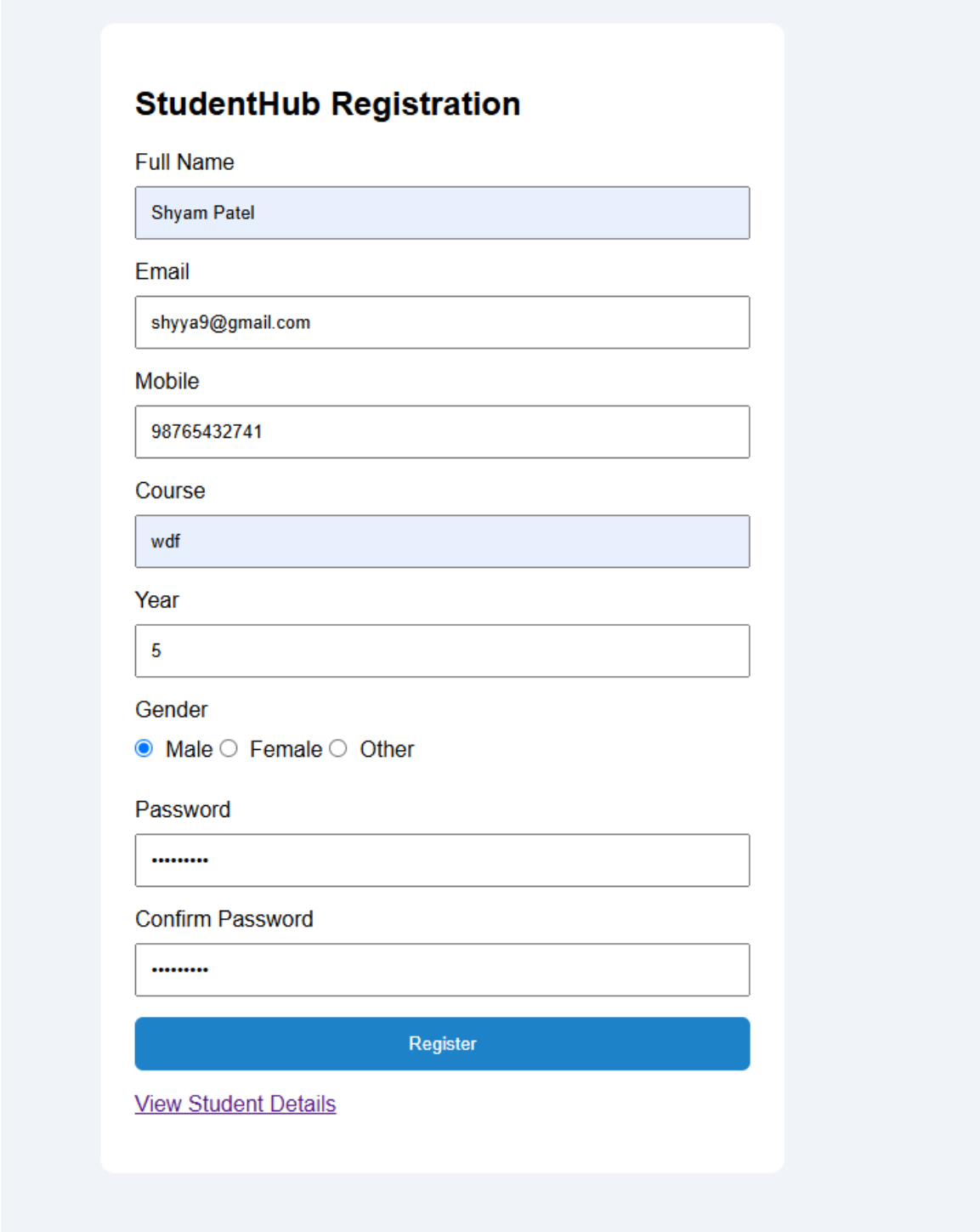
2Are primary keys and foreign keys correctly defined?

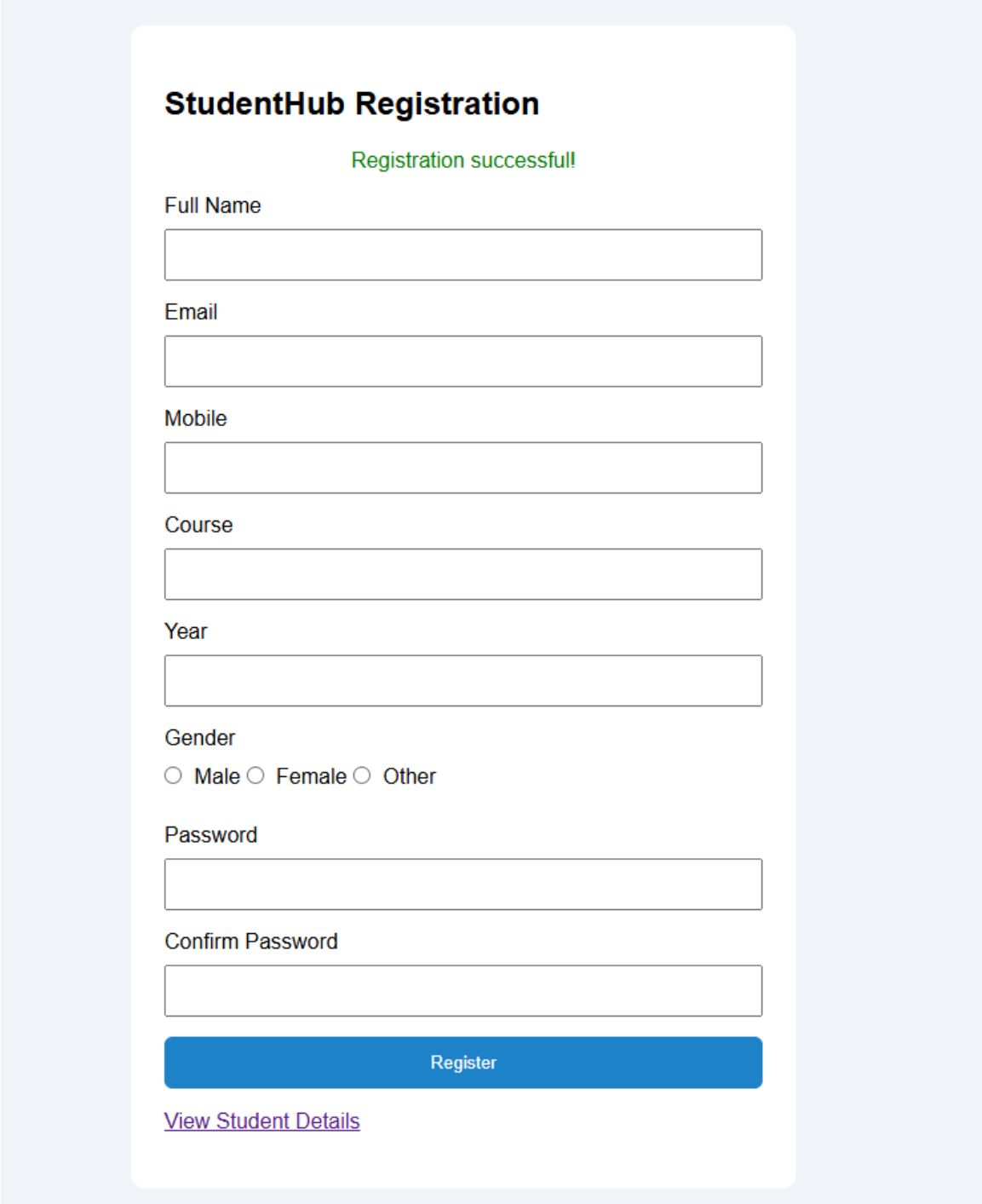
Yes. Each table has a **primary key**, and the Registrations table uses **foreign keys** to connect students and events correctly.

3. Is the database connection handled securely?

Yes. **PHP PDO and prepared statements** are used for secure database connection and to prevent **SQL injection**.

Implementation/ Output/Result


StudentHub Registration
Full Name
Shyam Patel
Email
shyya9@gmail.com
Mobile
98765432741
Course
wdf
Year
5
Gender
Male
Female
Other
Password
Confirm Password
Register
View Student Details


StudentHub Registration
Registration successful!
Full Name
Email
Mobile
Course
Year
Gender
Male
Female
Other
Password
Confirm Password
Register
View Student Details

StudentHub - Student Details

ID	Name	Email	Mobile	Course	Year	Gender
1	Shyam Patel	shyamkapuriya9@gmail.com	9876543210	wdf	2	Male
2	yash Patel	yash@gmail.com	9876543210	wdf	1	Female
3	Shyam	shyamkapuriya@gmail.com	9876543210	wdf	4	Male
4	Shyam Patel	shyam@gmail.com	9876543210	wdf	2	Male
5	Shyam Patel	s9@gmail.com	9876543210	wdf	2	Male
6	Satel	a9@gmail.com	9856232356	wdf	1	Male
7	Shyam Patel	shyya9@gmail.com	98765432741	wdf	5	Male

Add Another Student

	student_id	name	email	mobile	course	year	gender	password	role
Edit Copy Delete	1	Shyam Patel	shyamkapuriya9@gmail.com	9876543210	wdf	2	Male	$2y$10$bxamt/vzJmkSq3WJZY4NgumbDO3CYBCR9QSA1Fa8rzk...	student
Edit Copy Delete	2	yash Patel	yash@gmail.com	9876543210	wdf	1	Female	$2y$10$QP1bOWlAOup267VlNbMZ1uhKs.1OzxzxjyAHQLG8TKj...	student
Edit Copy Delete	3	Shyam	shyamkapuriya@gmail.com	9876543210	wdf	4	Male	$2y$10$eFPblzgrStfkg3LoyTA1K.00Mg3401irHnqGq.6jUpe...	student
Edit Copy Delete	4	Shyam Patel	shyam@gmail.com	9876543210	wdf	2	Male	$2y$10$3qboF8Rt12xliWn6IOAks.iHLkXDztHxcqg2.Zj2SiM...	student
Edit Copy Delete	5	Shyam Patel	s9@gmail.com	9876543210	wdf	2	Male	$2y$10$ixaNgOZgve4ff3pTPO4G4e26hxFwbYpyy4KLztpvqgV...	student
Edit Copy Delete	6	Satel	a9@gmail.com	9856232356	wdf	1	Male	$2y$10$52LwDYZwrXk9hwazCjwDfOkKj6SRg5QUwK9.6LoQBP0...	student
Edit Copy Delete	7	Shyam Patel	shyya9@gmail.com	98765432741	wdf	5	Male	$2y$10$oJfcrG1.CUg5Rz2w03P.kOl3raoSw5iH9K8vjLRyxP2...	student

https://github.com/ShyamPatel9/Student-Hub-Portal

Learning Outcomes:

I am learned how to design a normalized database using MySQL. They understood the use of primary keys and foreign keys to maintain data relationships. They also learned to connect PHP with MySQL using PDO. Prepared statements were used to improve security and prevent SQL injection.

Faculty Remarks: (Keep this blank for faculty)